AI-Powered Credit Risk Intelligence Platform

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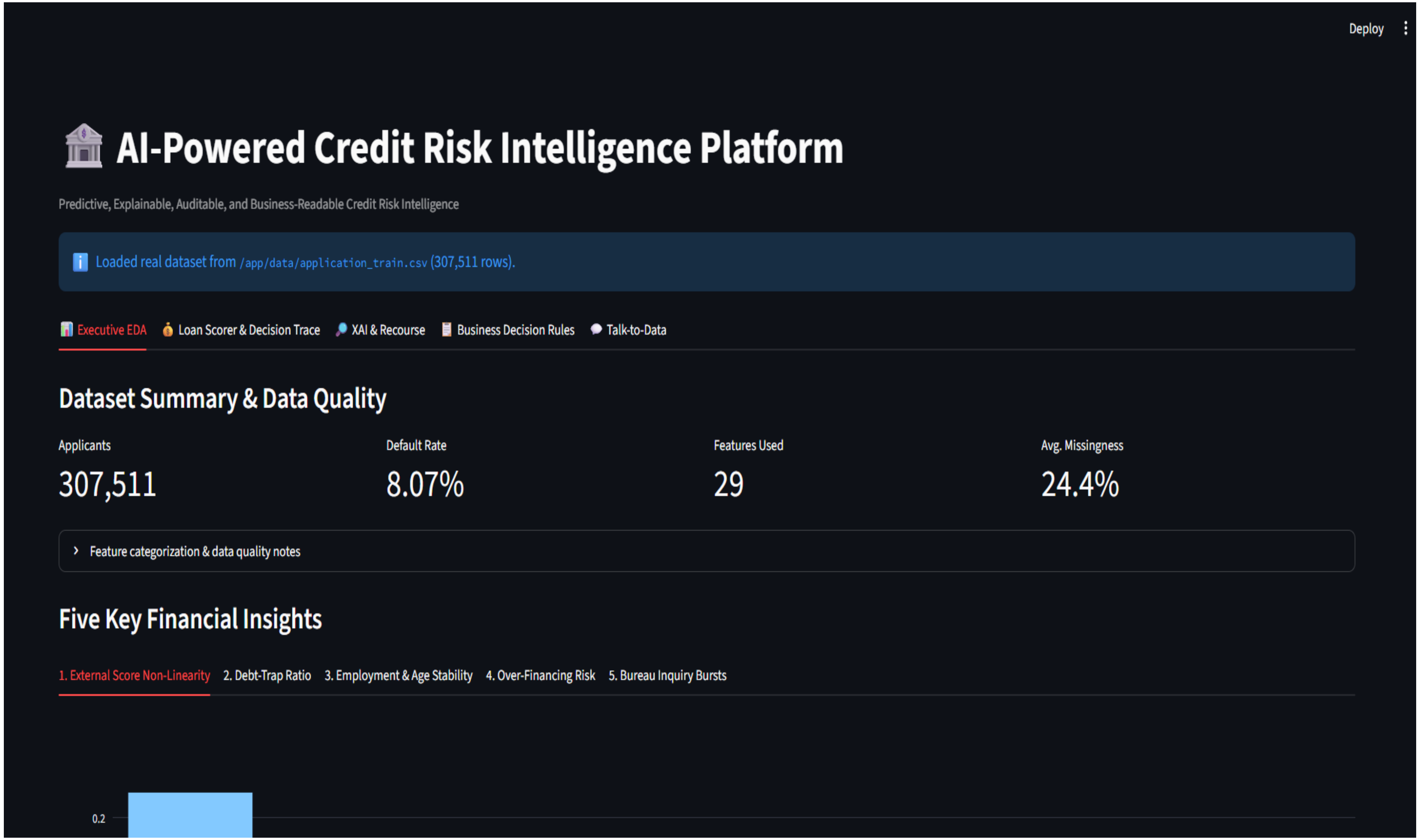
Problem Statement

- Credit institutions need to evaluate whether a loan applicant is likely to be a low-risk or high-risk borrower.
- **Proposed Solution**
- An **AI-powered Credit Risk Intelligence Platform** that analyzes applicant and financial information to:
- Predict credit risk
- Identify important risk factors
- Provide explainable decisions
- Support real-time loan assessment
- Assist business decision-making

Dashboard & Dataset Overview

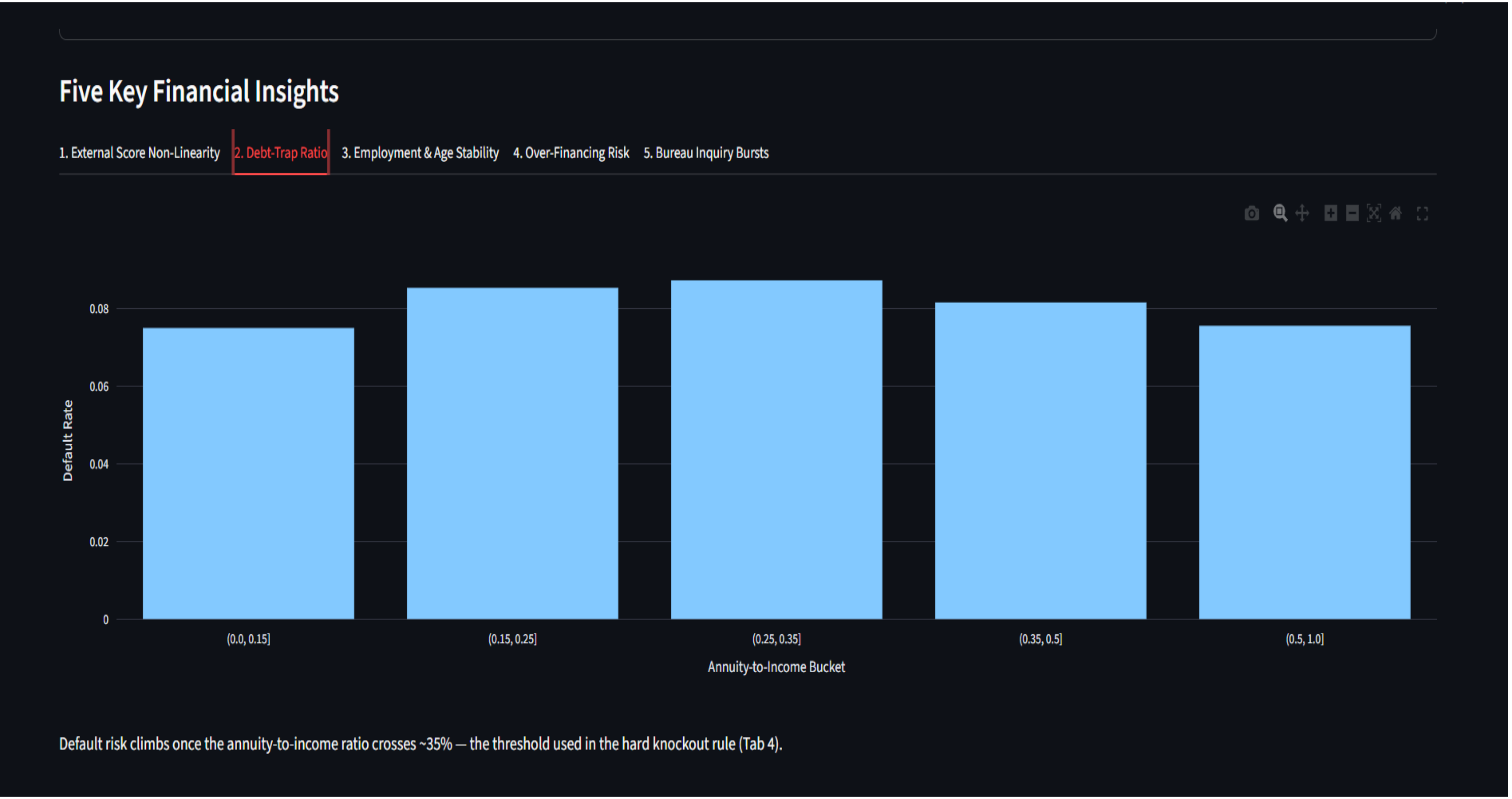
The dashboard is with an overview of the dataset and its quality. The dataset contains **307,511 applicants, 29 features, an 8.07% default rate, and approximately 24.4% average missingness,** providing the foundation for the subsequent analysis.



Financial insight – External score non linearity

The analysis shows that default risk changes sharply as the external bureau score increases rather than following a simple linear relationship. Applicants with lower external scores have substantially higher default rates, making this one of the strongest indicators of credit risk.

Financial insight – Debt-to-Income



The relationship between the annuity-to-income ratio and loan default risk. Default risk increases noticeably when the ratio crosses approximately **35%**, highlighting the importance of affordability and excessive repayment burden.

Financial insight – Employment and age Stability

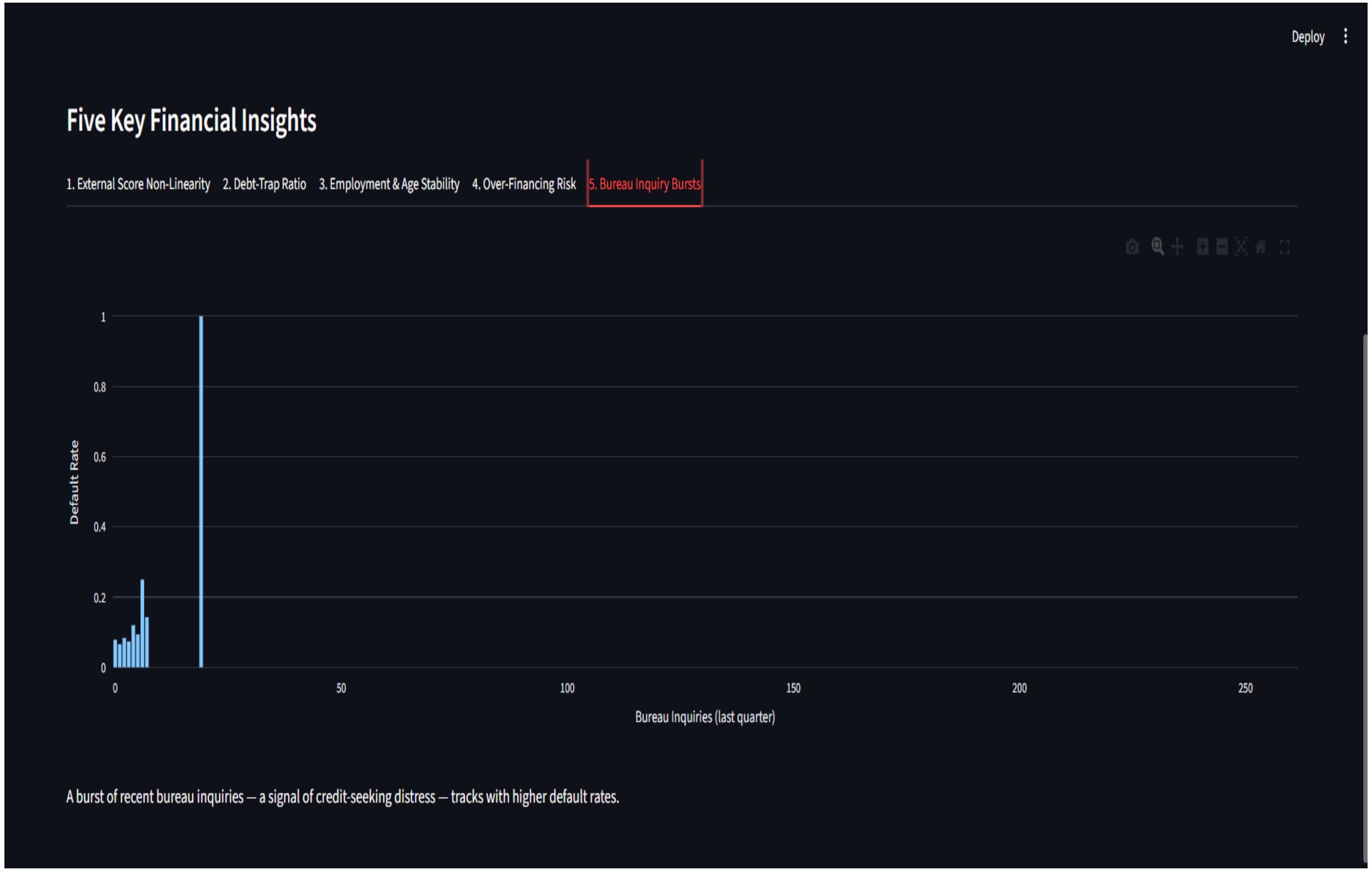
Age and employment stability are analyzed as indicators of borrower reliability. The results show that younger applicants, particularly those in the **20–30 age range**, have a materially higher default rate than older and more employment-stable applicants.

Financial insight – Over Financing risk

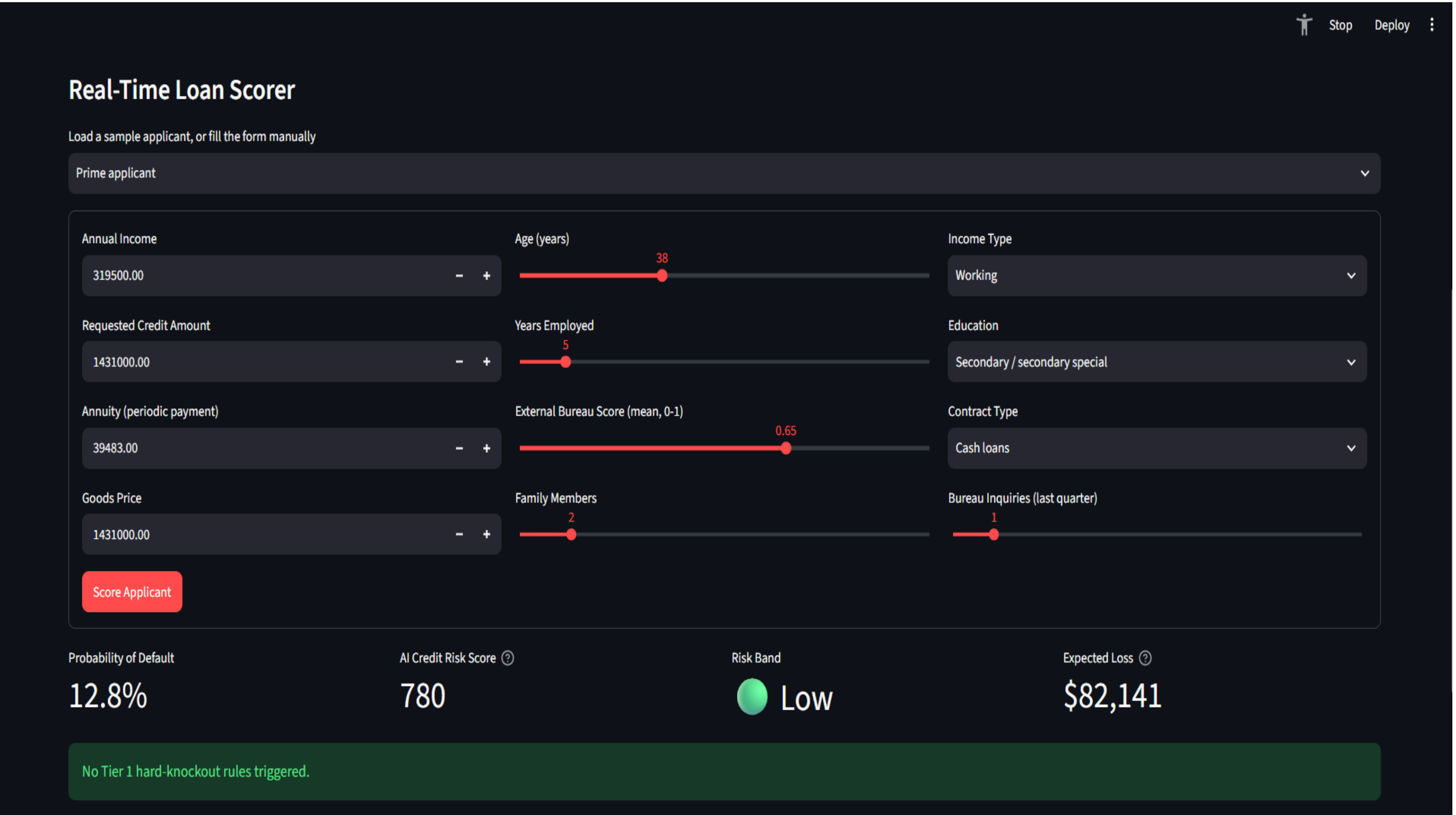
This analysis focuses on the relationship between the requested credit amount and the price of the purchased goods. Applicants requesting credit above the goods price demonstrate higher default risk, indicating potential over-financing or under-collateralized lending.

Financial insight – Bureau Inquiry Bursts

Recent bureau inquiries are used as an indicator of increased credit-seeking behavior. A higher number of inquiries within the last quarter is associated with greater default risk and can therefore act as an early warning signal.



Real Time Loan Scorer

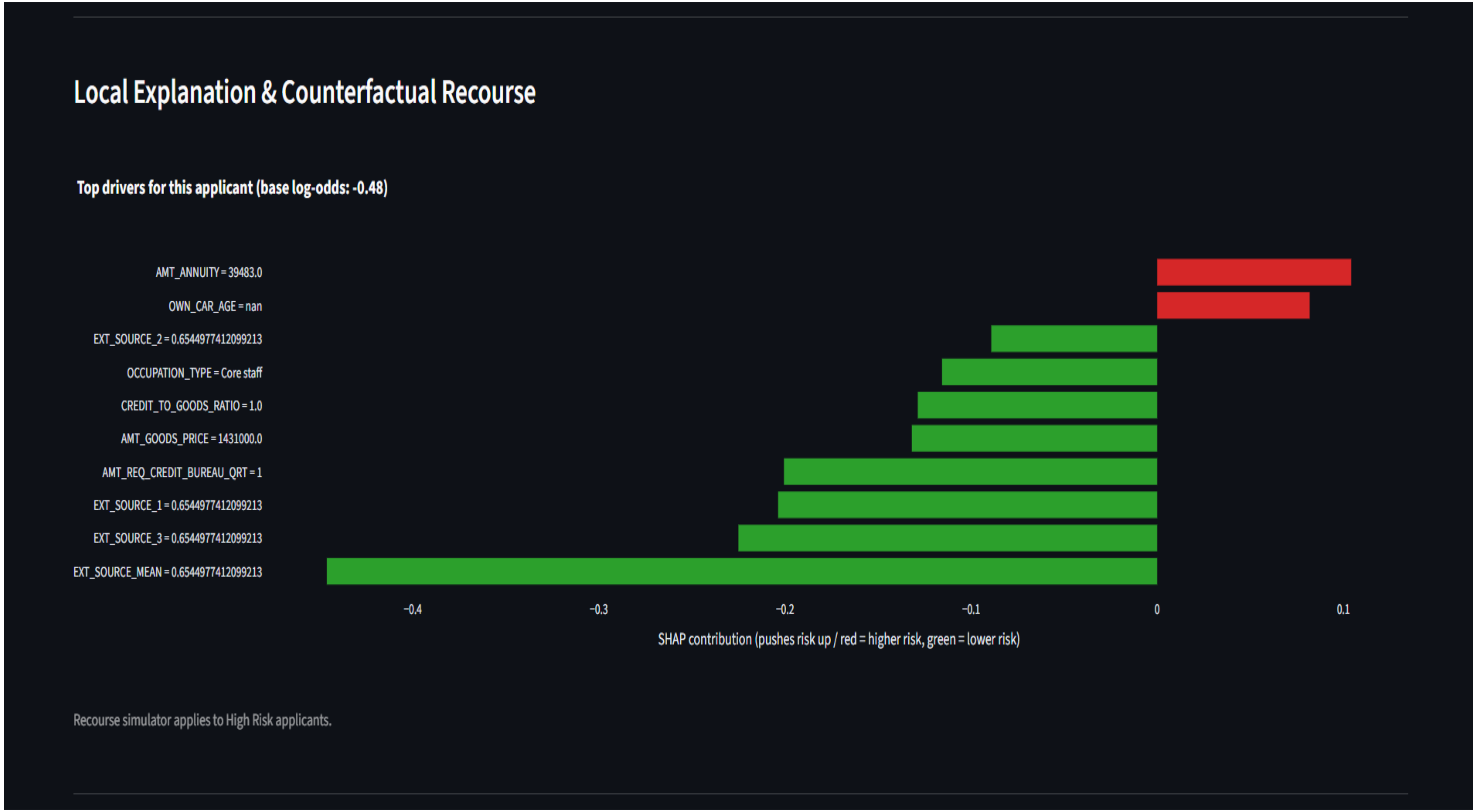


The dashboard converts the applicant's information into an actionable risk assessment. For the demonstrated applicant, the system reports a **12.8% probability of default, a Low risk band, an AI Credit Risk Score of 780, and an expected loss of $82,141**, while also checking policy rules.

Global Feature Importance

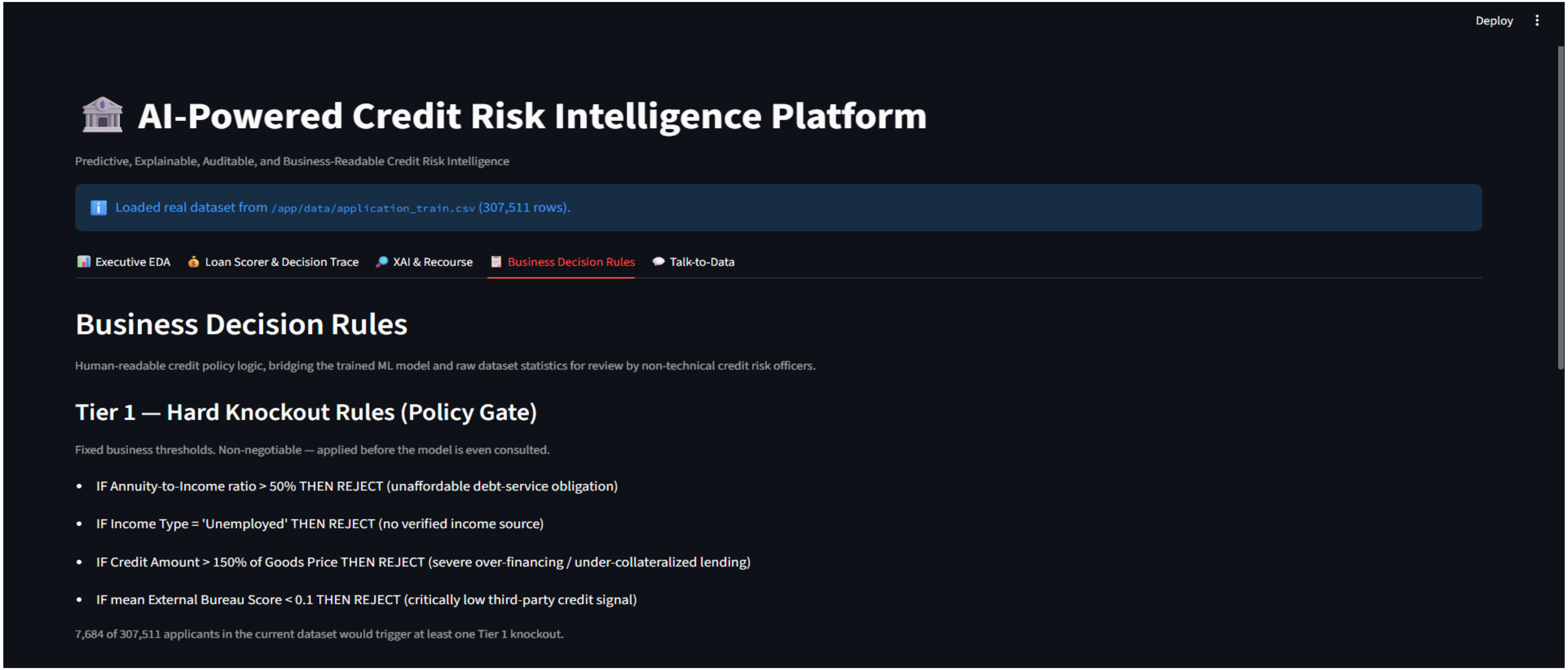
The dashboard shows which variables have the greatest overall influence on the model's predictions. Global SHAP feature importance provides transparency by showing the major risk drivers across the applicant population rather than focusing on only one individual.

Local Explanation & Counterfactual Recourse



The platform provides applicant-level explanations using SHAP contributions to show why a particular prediction was made. It also includes counterfactual recourse, allowing users to understand what changes to an applicant's characteristics could potentially improve their risk outcome.

Tier 1: Hard Knockout Business Rules



Tier 2: Model-Derived Business Rules

Tier 2 — Model-Derived Business Rules (surrogate decision tree)

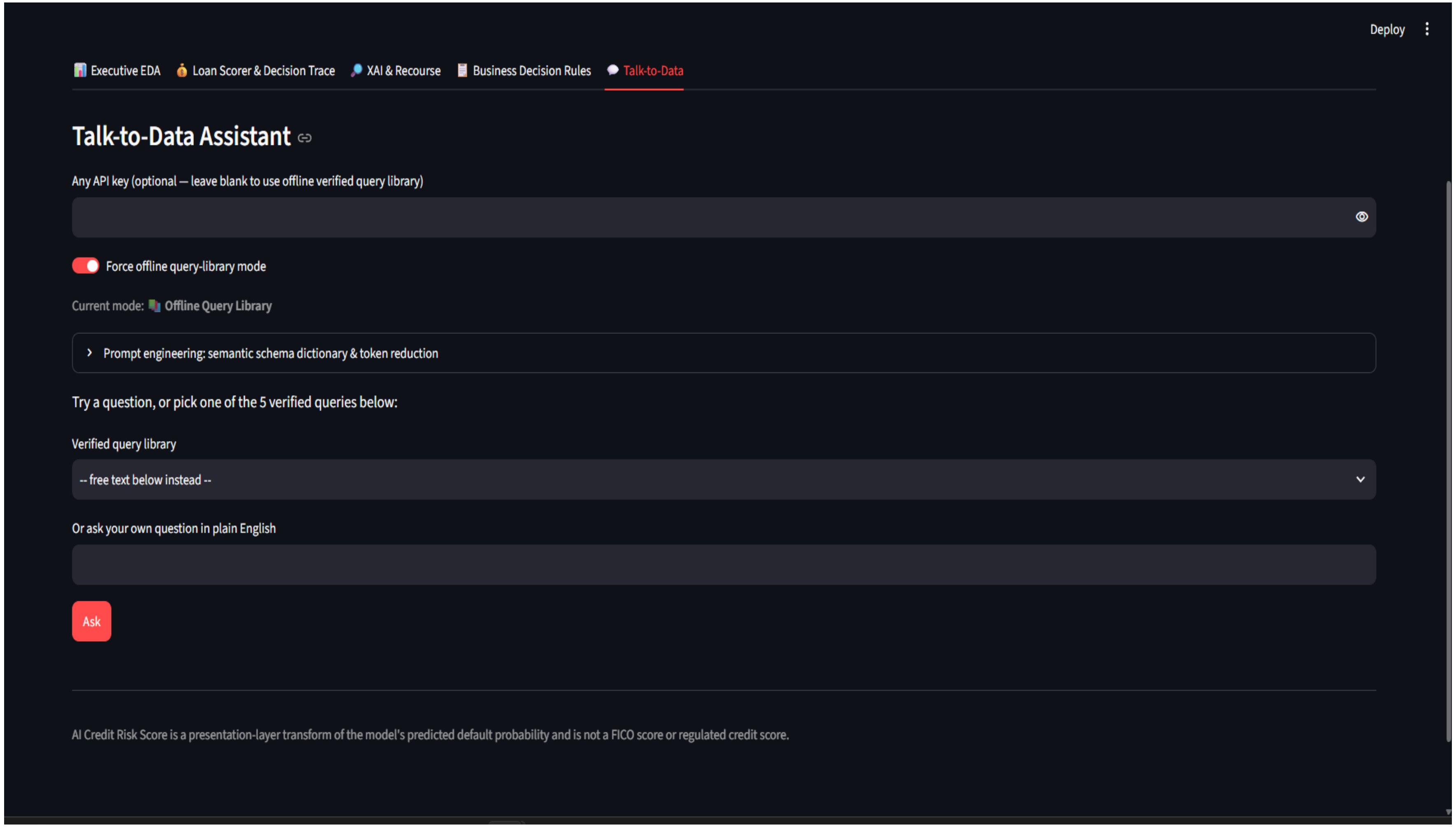
A shallow decision tree fit on the trained model's own High-Risk calls, read out as IF-THEN rules and audited against real outcomes.

View as:

- ( ) Plain-English bullets
- (•) Structured table
- ( ) Raw decision tree (sklearn.tree.export_text)

IF (condition)	Support (n)	Coverage (%)	Precision (%)
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN <= 0.25 AND EXT_SOURCE_3 <= 0.50 AND AGE_YEARS <= 59.66	1757	2.86	88.79
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN > 0.25 AND CREDIT_TO_GOODS_RATIO > 1.21 AND AGE_YEARS <	846	1.38	68.09
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN <= 0.25 AND EXT_SOURCE_3 > 0.50 AND AGE_YEARS <= 50.37	1054	1.71	63.47
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN > 0.25 AND CREDIT_TO_GOODS_RATIO <= 1.21 AND EXT_SOURC	647	1.05	50.85

Talk-to-Data Assistant



The Talk-to-Data Assistant provides a conversational interface for interacting with the credit-risk dataset. Users can select verified queries or ask questions in plain English, reducing the need to manually construct database queries or perform technical analysis.

Conclusion

- **Predictive:** Identifies applicants with higher probability of default.
- **Explainable:** Reveals the key factors behind every risk decision.
- **Actionable:** Combines AI predictions with business rules and real-time scoring.
- **Intelligent:** Enables natural-language interaction with credit data.
- **The platform transforms credit data into transparent, actionable intelligence for faster and smarter lending decisions.**

THANK YOU !!!!